

Falcon-VT1 Flight Controller

Design Document • Rev. A2 • Ayr Aerial Systems • Doc. No. DD-FVT1-A2

1. Overview

The Falcon-VT1 is a flight controller board for small VTOL fixed-wing UAVs, supporting up to 8 independent ESC channels, redundant IMUs, and a dedicated telemetry radio link. It is designed to drive the four thrust-generating devices used in tail-sitting VTOL transitions.

2. System Architecture

Figure 1 — Falcon-VT1 Flight Controller Block Diagram

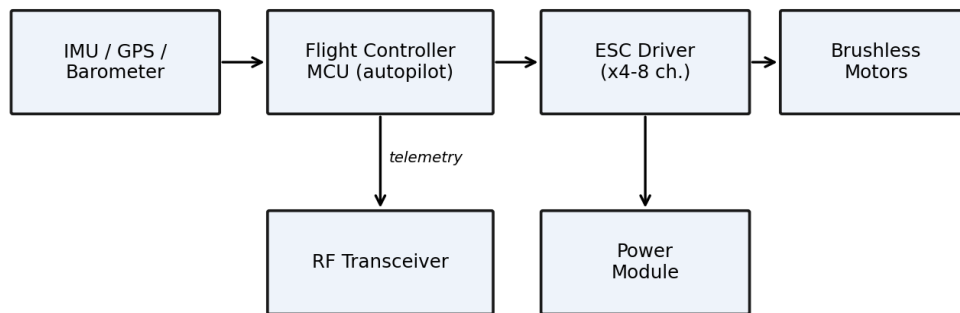


Figure 1 — Falcon-VT1 block diagram: sensor fusion, autopilot MCU, ESC driver bank, and RF telemetry.

The flight controller MCU fuses IMU, GPS, and barometric data at 400 Hz and outputs PWM/DShot commands to the ESC driver bank. The RF transceiver provides a bidirectional 915 MHz telemetry link to the ground control station for command uplink and status downlink.

3. Electrical Specifications

Parameter	Value	Notes
Input voltage range	9 - 26 V DC	2S-6S LiPo compatible
Max continuous current per ESC channel	40 A	8 channels
Peak current per ESC channel	60 A	10 s max
Processor	Dual-core Cortex-M7 @ 480 MHz	M4 co-processor
IMU update rate	400 Hz	triple-redundant IMU
Telemetry range	≥ 15 km	line-of-sight, 915 MHz
Operating temperature range	-20 °C to 55 °C	board ambient
Dimensions	68 x 46 x 14 mm	excluding connectors
Weight	38 g	board only

4. Known Limitations

The current revision (A2) does not provide hot-swap redundancy between the primary and secondary IMU; a failed primary IMU requires a reboot to switch. This is being addressed in Rev. B.